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ASIA CENTER

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Strategy for Sourcing, Structuring & Maintaining a Personalized Finance Knowledge Base

Team: **Minimum Viable Powerhouse** | Anuraag | Ekta | Gaurang | Jasjeev

Redefining Indian Personal Finance: How Mojek's youth-centric, AI-driven value proposition is solving the core PFM challenges & why over *70%* of young Indians are embracing simpler, smarter money management

How Mojek Money's Core Value Proposition is positioned to solve for the key challenges in the PFM Tools Market

Empowering youth to make smarter financial decisions

Delivering trusted, accurate & personalized finance solutions

Offering transparency, credibility, and ease of use

Strengths \ Challenges	Low financial literacy & manual expense tracking	Data security concerns & lack of real-time insights	Complex credit & tax planning	Limited investment knowledge & unstructured goals
Proprietary Categorization (60+ types)	✓			
Superior Bank Integration & Statement Analyzer		✓	✓	
Clean Minimal User Experience	✓		✓	✓
Agile & User Focused Dev Cycle				✓

Key User Personas

Key Attributes	Ambitious Young Professional (30 Yr.)	Side-Hustle Student (21 Yr.)	Digital-First Freelancer (26 Yr.)
			
Profile	IT Professional, steady salary, values security of data	Student balancing internship, juggling irregular income	Freelance Designer with inconsistent monthly income
Goals	Tracking & monthly budgeting, investing, tax management	Managing savings for higher education, learning about investments	Managing uneven inflows, taxes for freelancers, expense claims
Pain Points	Overwhelmed by financial jargon, limited tax knowledge	Unpredictable income stream, lack of investment literacy, needs easy expense tracking	Lack of access to traditional financial advice, prone to fraud transactions

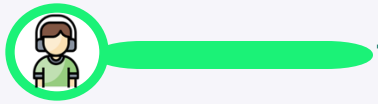
Personal Finance Software Market Landscape



4.6 CAGR%
Growth driven by digital payment infrastructure and rising financial literacy



64% Millennials



72% Gen Z

Believe digital tools make financial goals more achievable



83%

of Indian graduates show substantial trust on AI agents for financial advice

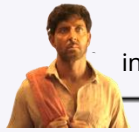
Market Landscape → **Needs Assessment** → **Top Questions** → **Knowledge Base** → **Data Storage** → **Retrieval Strategy** → **Governance** → **Risks & User Feedback Loop**

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Needs Assessment: Foundational, high-frequency pain activities, such as budgeting, expense tracking, and credit management, represent key moments of truth where a financial product must perform flawlessly to build user trust

How Influential the Need is

Low — High



Tracking all income sources

Automating expense categorization

Analysing spending patterns

Summarising transactions by merchant

Monitoring monthly budget goals

Downloading bank statements quickly

Managing credit card due dates

Setting specific savings goals

Getting personalized tips on savings

Setting up recurring investments

Visualizing cash flow trends

Learning financial concepts

Estimating the future cost of goals

Monitoring your CIBIL score

Analysing your credit report

Receiving alerts for suspicious transactions

Discovering tax-saving instruments

Planning your taxes

Strategizing to pay down debt

Planning for major life goals

Managing all your insurance policies

Understanding your risk appetite

Tagging potentially tax-deductible expenses

Spotting hidden fees fast

Comparing your spending and saving habits

Identifying all your recurring payments and subscriptions

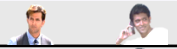
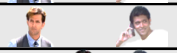
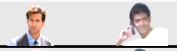
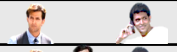
Tracking collective financial goals



Most Important Need Categories

1. **Budgeting & Expenses**
   - Frequency: High
   - Urgency: High
2. **Credit & Borrowing**
   - Frequency: High
   - Urgency: High
3. **Investing & Market**
   - Frequency: High
   - Urgency: High
4. **Taxation**
   - Frequency: High
   - Urgency: Medium
5. **Fraud & Security**
   - Frequency: Medium
   - Urgency: High
6. **Credit Health**
   - Frequency: Medium
   - Urgency: Medium
7. **Investment Discovery**
   - Frequency: Medium
   - Urgency: Medium
8. **Long-Term Goal Planning**
   - Frequency: Medium
   - Urgency: Medium
9. **Cash Flow Forecasting**
   - Frequency: Medium
   - Urgency: Medium
10. **Debt Reduction**
    - Frequency: Medium
    - Urgency: Medium

Top Questions: For the top needs categories, we must focus on high-criticality questions that cater to many personas to maximize user value before developing more complex advisory tools

Need Category	Criticality	Questions	Personas	Question Score
01. Budgeting and Expense Management		"How much did I spend on [category] this month (e.g., dining, groceries)?"		★★★
		"Notify me if I exceed my daily/weekly budget of ₹X."		★★★
		"Suggest ways to save ₹Y per month based on my spending patterns."		★★★
		"Compare last month vs. this month's expenses and highlight variances."		★★
02. Credit and Borrowing		"What's my current CIBIL score, and how can I improve it?"		★★★
		"Am I eligible for a ₹30L home loan with a 7% interest rate?"		★★
		"Recommend the best credit card for cashback on groceries given my spending."		★★
		"Calculate EMI for Bank Z's ₹10L personal loan offer of 12% over 5 years."		★
03. Investing and Market Insights		"Should I choose SIP or lump-sum for a ₹10K investment this month based on my portfolio?"		★★★
		"Show top 3 mutual funds by 3-year returns in the ELSS category."		★★
		"What are current FD, digital gold, and gold-ETF rates?"		★★
		"What's my portfolio's equity allocation drifts over time?"		★
04. Taxation and Regulatory Updates		"What's the deadline for ITR filing this year, and how do I file online?"		★★★
		"Explain new Section 80C limits and eligible expense categories."		★★
		"Calculate capital gains tax for my ₹1L mutual-fund redemption after 2 years."		★
		"What SEBI guidelines were issued on crypto exchanges this quarter?"		★
05. Fraud and Security Alerts		"Alert me if a transaction over ₹1,000 is made outside my usual locations."		★★★
		"Can you check if *PayQuick* has been flagged for fraud?"		★
		"Have there been any suspicious transactions in my account in the past week?"		★★
		"What are the best practices to secure my digital wallets?"		★

 Ambitious Young Professional

 Side-Hustle Student

 Digital-First Freelancer

Data Sources: 30+ data sources are required for the top data categories to accurately answer the top questions of the target users, with each data source having varying frequencies of retrial and refresh

MONETARY & MACRO ECONOMIC

- RBI Official Data Repository
- MoSPI CPI
- RBI Forex Reference Rates
- RBI Credit-Deposit Rates

MARKET DATA & SECURITY PRICES

- AMFI NAVs & Scheme Info
- BSE Data Feed
- NSE Data Feed
- CAMS Data Services

PAYMENTS & TRANSACTION DATA

- User bank transactions
- User UPI payment history

FRAUD PREVENTION

- Datarade Fraud Detection Datasets

TAXES & REGULATORY DATA

- Income Tax Analyzer with IT department API*
- SEBI Circular Tracker

SOCIAL WELFARE & INSURANCE

- ESIC Contribution Rates
- IRDAI Reports
- PFRDA NPS Scheme Info

UTILITY & RETAIL DATA

- Amazon/Flipkart Catalog
- Petrol & Diesel Price APIs
- State Electricity Tariffs

USER PERSONAL DATA

- User profile
- User financial goals
- User expense categorization
- User investment portfolio
- Loan information**

CREDIT SCORE & RISK DATA

- TransUnion CIBIL Score API
- Equifax Credit Data API

15+ Common User Queries in just 5 question categories

30+ Common Data Sources to answer just the above questions

∞ Rules and Regulations in the Indian finance sector

Critical need for a **user-friendly** approach to answer these questions



* Mainly external information since user specific tax data requires additional consent

** Coming soon

AI CHAT BOT

Have there been any suspicious transactions in my account in the past week?

Show top 3 mutual funds by 3-year returns in the ELSS category.

Alert me if a transaction over ₹1,000 is made outside my usual locations.

What are current FD, digital ETF rates?"

What's my portfolio's equity allocation drifts over time?

Can you check if PayQuick has been flagged for fraud?

What SEBI guidelines were issued on crypto exchanges this quarter?

"How much did I spend on [category] this month (e.g., dining, groceries)?"

Notify me if I exceed my daily/weekly budget of ₹X."

Calculate capital gains tax for my ₹1L mutual-fund redemption after 2 years.

What are the best practices to secure my digital wallets?

Compare last month vs. this month's expenses and highlight variances.

Suggest ways to save ₹Y per month based on my spending patterns.

We just got in our fresh a today!

What's my current CIBIL score, and how can I improve it?

Explain new Section 80C limits and eligible expense categories.

Am I eligible for a ₹30L home loan with a 7% interest rate?"

What's the deadline for ITR filing this year, ar file online?

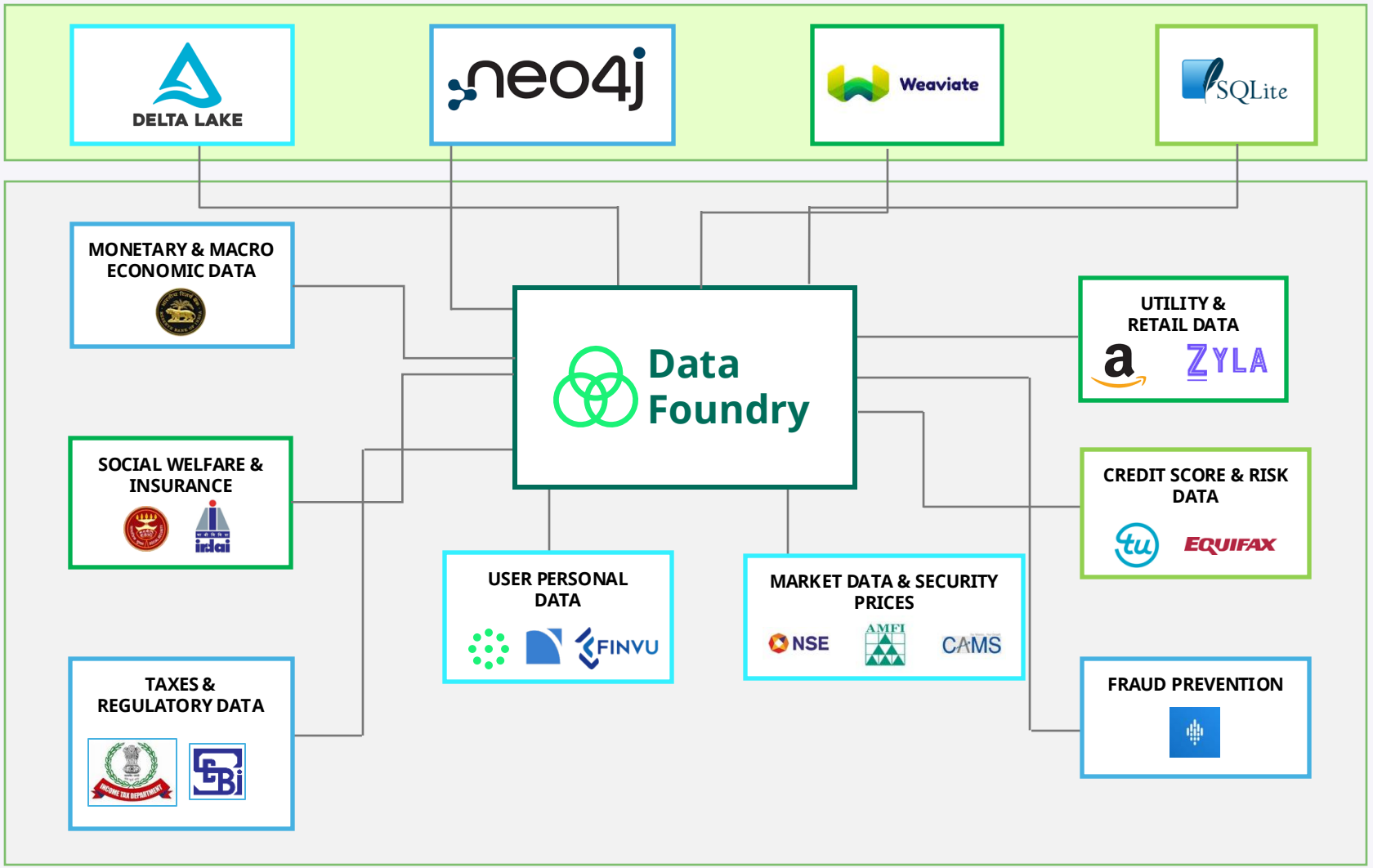
We just got in our fresh autumn collection today!

Calculate EMI for Bank Z's ₹10L personal loan offer of 12% over 5 years.

Recommend the best credit card for cashback on groceries given my spending.

"How much did I spend c (e.g., dining, groceries)?"

Data Storage: A "Data Foundry" architecture, which uses a sophisticated hybrid data core, orchestrates how the data is stored and retrieved to deliver personalized insights



THE PROBLEM

The Insight to Action Gap

- With 30+ siloed data sources our ability to deliver the hyper-personalized answers our users demand. This friction in the user journey prevents us from converting data assets into actionable insights, failing to capture ultimate value.

THE FOUNDATION SOLUTION

Moving from Silos to a Unified Knowledge Core

- For this we engineer a unified knowledge core. By centralizing our disparate data sources into a single, reliable asset, we have established the prerequisite for delivering personalization at scale.

THE FUTURE

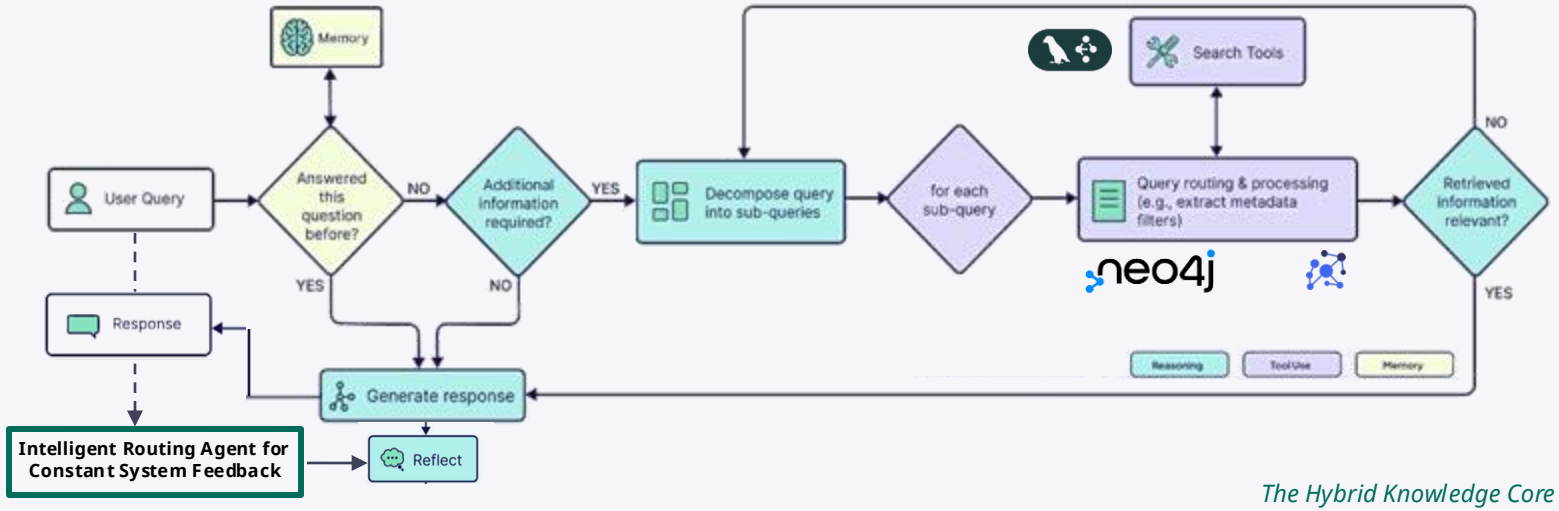
Activating Intelligence

- Go beyond just unification into actionable insight – decomposing complexities to bite sized answers

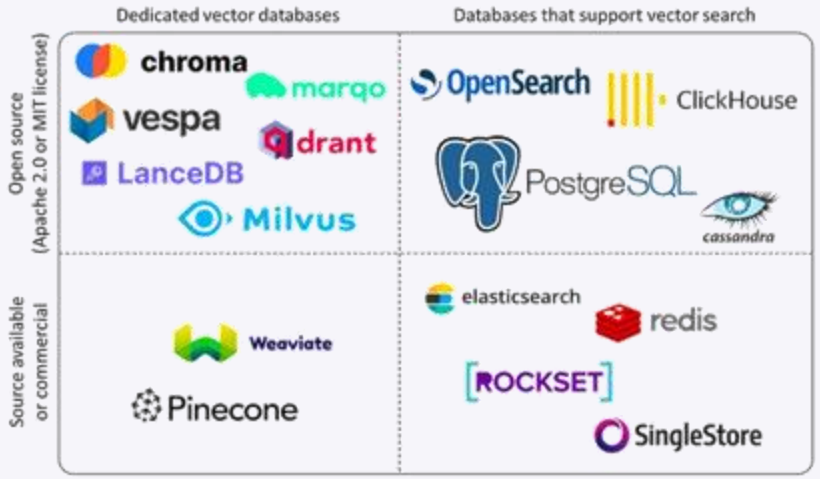
Simplifying Chaos with the Hybrid Data Core



A **Hybrid Agentic Architecture** for delivering grounded financial advice, moving beyond the simple searches fusing the knowledge core with user data



The Hybrid Knowledge Core



Comparison - Vector DB Landscape

HYBRID KNOWLEDGE CORE

Knowledge Graph for structured facts and rules

***Vector DB** for **semantic search** over unstructured text (e.g., market analysis, news)*

***Fuse sparse (keyword-based) and dense** (semantic) **search** results using Reciprocal Rank Fusion (RRF)*

LangGraph specializes in creating **controllable, stateful agents** required for our use case.



I read two articles online about capital gains tax that contradicted each other. I'm terrified of making a mistake and getting a notice from the IT department.



I asked my bank's bot to compare a tax-saving FD with an ELSS fund, and it just sent me a generic link to their investment page

80%

Users abandon sites due to poor search XP*

5 Out of Every **6**

Young Indian investors express significant trust in receiving financial advice from AI –tools**

Optimized for Mobile-First with all heavy computation on the server to **increase inference speed by 2-3x**

A *Query Decomposition* agent splits complex questions into sub-queries, which a *Routing Agent then dispatches for multi-step reasoning* - **boosts accuracy on complex reasoning tasks by ~30% over rule-engines** (Google)

*Source : *(Nosto'23) | **(CFA Institute'24)*

Governance Framework: Successful implementation requires prioritizing data maintenance efforts, building a data governance architecture, and enforcing the proposed RACI model

Why: The Needs of Governance Architecture

- Ingest data from **numerous and diverse sources.**
- Enforce a **consent and privacy framework** and **data protection regulations**.
- Execute data pipelines as **scheduled batch runs** at regular, user-defined intervals.
- Implement automated **validation checks,** ensuring integrity and quality of data.
- Provide **centralized, continuous monitoring** of all data pipelines.



How: Data Governance Architecture

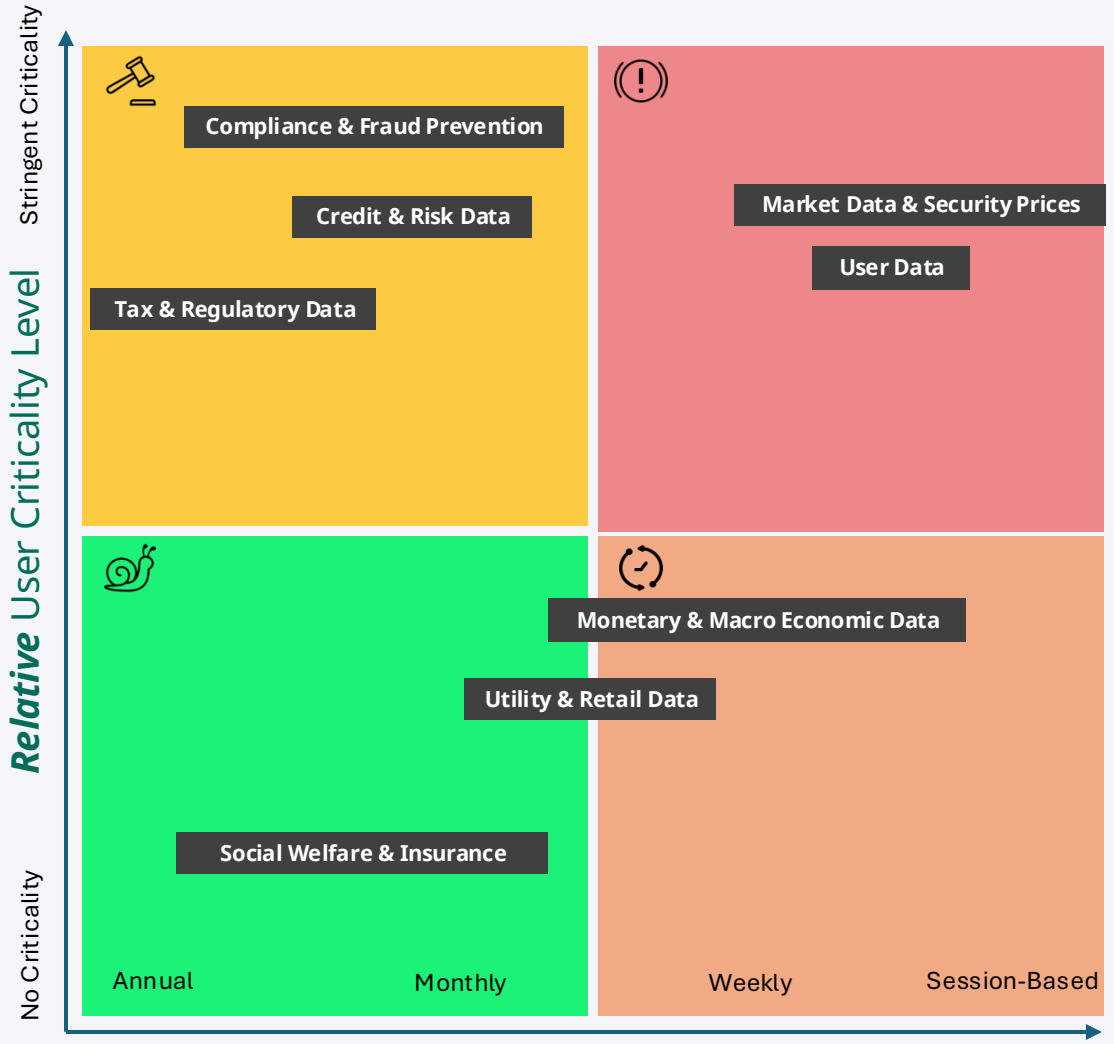
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Treats data as trackable assets, integrating automated quality checks, and using a flexible scheduler to handle the varied update frequencies your system requires.

Who: RACI for Key Stakeholders

Activity / Process	Executive Sponsor	Data Governance Lead	Data Quality Analysts	Legal & Compliance
Define Data Quality Strategy	A	R	I	C
Implement Automated Source Monitoring	A	R	R	C
Schedule Source Refresh	A	R	I	C
Metadata & Version Control	A	R	I	C
Consent & Privacy Compliance	A	C	I	R
Security & Access Control Policy	A	C	I	C
Incident & Escalation Management	A	R	R	C
Policy Review & Update	A	R	I	C

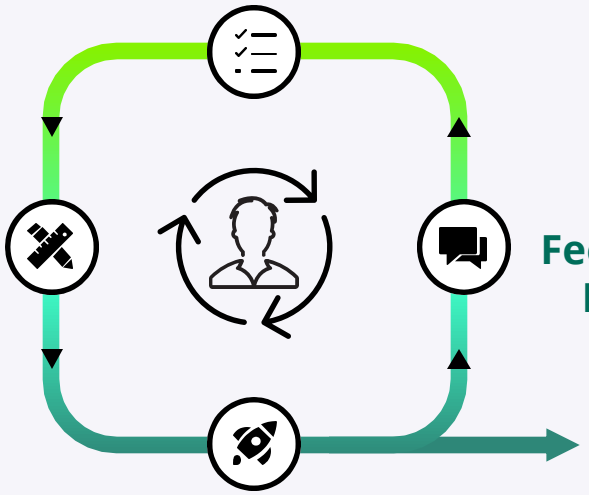
What & When: Data Maintenance Matrix



Market Landscape | Needs Assessment | Top Questions | Knowledge Base | Data Storage | Retrieval Strategy | **Governance** | Risks & User Feedback Loop

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User Feedback Loop & Risk Management: A holistic, continuous improvement architecture leverages a zero-latency user feedback loop as the primary defense to proactively mitigate the platform's most critical risks



User Feedback Loops



Immutable Trail - Burn feedback into a versioned Neo4j graph creating a permanent, auditable trail, social listening for insights.

Identify patterns in user corrections (e.g., repeated confusion between ELSS and ULIP tax benefits) and uses them to automatically retrain the Intelligent Routing Agent

Zero Latency Loop - Feedback is captured asynchronously via a message queue, so it never slows the user down. Parsed and logged post a malice check

Risks of Failure

Poor User Trust Due to Incorrect Advice

Risk: If the AI delivers inaccurate or conflicting guidance, users will rapidly lose trust and abandon the platform.

Mitigation: Multi-layer validation for answers and provide seamless human escalation for unresolved queries.

Non-Compliance with Regulation

Risk: Inability to keep pace with evolving financial rules may result in heavy fines, investigations, or even a ban.

Mitigation: Make compliance a central governance focus using dedicated experts and automated policy tracking.

Cybersecurity Breaches

Risk: Data leaks or hacking incidents can cause catastrophic user harm and brand destruction.

Mitigation: Invest in top-tier cybersecurity, routine penetration testing, and rapid incident mitigation protocols.

Risks of Success

Amplification of Market Trends by Uniform AI Advice

Risk: If most users act on the same automated AI advice, it may unintentionally increase market volatility or lead to systemic risks.

Mitigation*:* Employ diverse model ensembles and monitor for crowding effects to reduce systemic risk.

Dependency on Third-Party AI/Cloud Providers

Risk: Heavy reliance on a single AI or cloud vendor could cause outages or security incidents affecting all users.

Mitigation*:* Use multi-cloud architecture, build redundancy, and have disaster recovery plans in place.

Ethical and Bias Risks

Risk: Automated financial guidance might reinforce existing socioeconomic biases or systematically exclude certain users.

Mitigation: Conduct ongoing AI fairness/ethics audits, audit datasets for representation, and collect diverse user feedback.

Thank You

Appendix | Data Sources to Data Type

Data Source	Storage/Categorization	Database/Solution	Comments
RBI Official Data Repository	Structured, Time-Series	Delta Lake (Parquet on S3, Delta)	Daily/weekly batch ingestion; fast analytics, history tracking
MoSPI CPI	Structured, Time-Series	Delta Lake	Monthly batch, historical trend analysis
RBI Forex & Deposit Rates	Structured, Time-Series	Delta Lake	Daily/Weekly ingestion
AMFI NAVs & Scheme Info	Structured, Time-Series	Delta Lake	Daily text file to Delta conversion, partitioned by scheme/date
SEBI Circular Tracker	Unstructured (PDF)/Regulatory	Delta Lake + Vector DB (Pinecone)	OCR extraction; semantic search & retrieval
IRDAI Reports, ESIC, PFRDA	Unstructured (PDF/HTML), Reports	Delta Lake + Vector DB (Pinecone)	Batch download, extracted and indexed
State Electricity Tariffs / Fuel APIs	Structured, Tabular, Regulatory	Delta Lake	Synchronized with other macro data
FIU-IND Fraud Alerts	Structured + Event Log	Delta Lake + Neo4j	Batch load, append-only; alerts graph-modelled for relationships
User bank transactions / UPI history	Structured, Session & Historical	PostgreSQL + Delta Lake	Latest session: Redis; full archive: Delta/Parquet
User credit scores, loan schedules	Structured, Secure, User-specific	PostgreSQL	Session retrieval from API, archived if user consents
User portfolio	Structured, Portfolio-State, Hybrid	PostgreSQL + Delta Lake	Persistent: PostgreSQL; batch snapshots to Delta Lake for analytics
User expense categorization	Structured, User-centric	PostgreSQL + Delta Lake	Categories in RDBMS; aggregate analytics in data lake
User tax filings	Structured/Unstructured, Secure	PostgreSQL + Delta Lake	Secure doc store, batch summary into lake
Amazon/Flipkart Catalog	Unstructured, External API	Not stored (API/Session only)	On-demand; ephemeral cache in Redis if required
BSE/NSE/CAMS Feeds	Structured, Time-Series/Events	Delta Lake, Redis (for session)	Delayed data stored; on-demand quotes session-cached
Setu FASTag Integration	Structured, Events, Session-based	PostgreSQL (user-level), Delta Lake (aggregate)	Events to user session, batch persisted
Income Tax Analyzer with IT Dept.	Structured, Secure	Session cache only (Redis), PostgreSQL for authorization logs	No user data retained except flow metadata
TransUnion CIBIL/Equifax	Structured, User/Credit Data	No storage/user session cache (Redis)	Consent required per query
User profile/personal/goal data	Structured, Profiled, Persistent	PostgreSQL	Operational tables
Market research/unstructured docs	Unstructured, External	Vector DB (Pinecone/Weaviate)	Indexed, searchable on semantic content
Regulatory and legal relationships	Graph (semantic/relationship)	Neo4j	Product, compliance, or actor relationships modelling
Session data (intermediate/user context)	Session, Temporary Cache	Redis	Cleared after session/logged in RDBMS as needed